

ActInf GuestStream 073.1 ~ Lachlan Kent: "Mental gravity: Modelling the embodied self"

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hello and welcome it is March 3rd 2024 at least where I am and we're here in active guest stream number 73.1 with Lan Kent mental gravity and I guess all that comes along with it so thank you for joining to you for a presentation and then we'll have some questions from live chat so thank you and to you thanks Danel my name's Lan Kent I'm here in Melbourne it's Monday morning at 6:00 a.m. so I work at UH rmit at sort of Tech University here where I do very applied uh practical Research into mental health promotion and prevention and I think one of the ma major barriers to progress in my field is that we don't have a really solid scientific theory of what constitutes mental health and well-being and by contrast how to Define mental ill health for the most common conditions like anxiety and depression so I've been working for a number of years on a theory of mental well-being effectively based on the foundational and Universal experience of gravity to try and get to the root explanation of why positive and negative mental states have a kind of vertical Dimension to them so depression for example is a painful down state anxiety is a painful up State well joy is an up state as well but it's Pleasant not painful and mindfulness or calmness are Pleasant down States so I think there's kind of explanatory power in those distinctions uh when we look at how gravity shapes biology cognition and phenomenology so I call it mental gravity but I don't mean anything mysterious by it just a way of modeling sort of physiological mental and emotional states on their physical Roots right next slide thanks Daniel before I start though I'd like to acknowledge the unseed lands of the bonang people of the Eastern colon Nation where I'm presenting from today and like respectfully acknowledge their Elders past and present and next slide again okay so today's talk we're going to go through a few sections we're going to start with a nice accessible anecdote a parable we're going to talk about Barbie uh the Barbie film and Barbie's fall and we're going to give a quick overview of the theory itself and then really focus on one of the key Concepts and controversial Concepts I think for mental Gravity the idea of a center of mental gravity and then relate that to active inference sort of give a summary of Where I

Come From and then take questions okay so let's start with a kind of parable of mental gravity Barbie's fall in the recent uh film starring Australian Margo Robbie so maybe you've seen it but even if you haven't it's not a very uh complex premise for the film so here we are at the start of the film and Barbie sort of coasting along in her usual Happy mode so Life's good she has a strong sense of self and the world is inherently predictable she's happy and this is depicted in the film as as her um floating kind of joyously sort of light and carefree but uh next slide thanks Daniel but then the whole narrative Arc of the film revolves around her fall she starts uniformly positive but then there's no spoilers this is in the trailer she experiences an identity crisis so instead of being happy all the time she starts feeling negative emotions something unpredictable happens that causes her to lose her sense of self and on account of the loss she experiences kind of an emotional breakdown she says this is the lowest of ever been emotionally and physically if you could just click once there and to really nail home uh the metaphor when Barbie Falls these are the lyrics of the really beautiful melancholic song by Billy eyh it's I used to float now I just fall down I used to know but I'm not sure now what I was made for what was I made for so it's an existential crisis for her that goes to the the core of her being she's not just sad because she lost something that can be replaced or even someone that like dear to her she kind of lost herself and when people get depressed or have an identity crisis like this we often say things like they're going through a mental breakdown so depression affects hundreds of millions of people each year around the globe it's kind of a universal phenomenon that needs scientific explanation but we still really have no idea what we're talking about really I think there's an intrinsic meaning here in the Fallen status of depression they can fulfill a sort of much needed scientific explanation you just click through to the next slide Daniel please okay now here's a spoiler although you won't be surprised Barbie ends up kind of happy again at the end she goes from really high so it's sort of Defying Gravity to low where she's overwhelmed by gravity gravity to somewhere kind of in between that's more balanced and more grounded where she's kind of managing gravity better so both the song and the film end on a hopeful note so Barbie recovers from her fall when she rediscovers a new sense of self and the song Now goes I think I forgot how to be happy it's something I'm not but something I can be it's something I wait for something I'm made for so mental gravity to me is not just about falling it's also about what people are for and what life is for so we have to have ups and Downs they are a necessary part of life but we all want the positives to sort of outweigh the negatives that's another gravity based metaphor we want to be up more than we're down so mental gravity is also about how to recover from Falls or maybe

how to prevent injuries from falling like depression so it's really a theory of mental health and well-being of emotional positivity of sort of human value of what we're made for we just click through to the next slide Daniel thank you okay so that's the pop psychology version now for the proper scientific explanation next slide things right so far in the last sort of 12 or six or 12 months I've uh published two papers on mental Gravity the first of which was pretty Technical and focused solely on depression the second of which is shown here it's a bit less Technical and more broadly focused on positive and negative Mental Health so it was in the Journal of environmental psychology and as you can see from the figure or hopefully you can see in the figure it framed the theory in terms of external environments at the top internal environments at the bottom and kind of intermediate socio emotional environments in um in the middle so we're going to take a quick tour through this summary figure and let's kind of start not at the top but in the middle in the social emotional environment where we found Barbie at a low sort of mental and physical emotional state you just click through a few times there okay and then next slide okay so the central idea in mental gravity is that there's a Continuum of emotional states that range from up high and ecstatic to fallen down and depressed so it's not a binary of course verticality isn't always positive there are sort of intermediate States so being up can also be a negative if you're unbalanced and say anxious about falling and being down can feel good if you're balanced stable and calm like in mindfulness meditation or just regular relaxation so these emotional states are a combination of physical States derived from complex interactions with the external environment and mental States derive from similarly complex reflective interactions within the internal environment so there's a kind of mapping between physical and mental states that mental gravity seeks to explain so assuming this is a human Universal throughout cultures and evolution how does this mapping come about what are the neural structures that regulate it and what does it mean to have gravity in or of the Mind next slide thanks Daniel okay so let's start with where it comes from so the principle underpinning mental gravity is called cognitive gravitropism now gravatropism in plants and animals is the ability and kind of the drive to maintain alignment to and elevation within the gravitational environment so for example plants grow vertically animals have muscular skeletal structures to maintain verticality and balance fish float using air sacs Birds fly with wings Etc so these structures and functions that regulate verticality have positive adaptive value for survival and evolution so maintaining gravitational uprightness and stability is a homeostatic function like temperature control or pH or nutrition whatever so if we fail to regulate our body's interaction with gravity we die it's as simple as that now cognitive gravitropism is

where that positive adaptive and homeostatic value has been internalized and co-opted as a kind of mental template for Value in other domains namely socio emotional expression or experience it's kind of the value of the self in socio emotional terms so this template means that we think talk feel and act in ways that are also gravitropic because being mentally up and aligned is generally preferable to being down and misaligned so this effectively means that the mind has a vertical Dimension that people strive to enhance through socio emotional structures and functions thoughts behaviors and emotional Expressions kind of all act in service of a drive to optimize mental verticality so this shows up in languages metaphor in English happiness is up unhappiness is down Consciousness is up unconsciousness is down health is up ill health is down more is up less is down so cognitive gravitropism is what makes these metaphors meaningful and kind of universal because they also appear in other cultures and language and languages and even in preverbal infants so mental gravity predates language in development at least and next slide thanks Daniel which brings us to where uh gravitropism comes from so we're at the top we're in the external environment so it's interactions with the external physical environment during childhood development so gravity has been a constant through human evolution it hasn't changed but its effects are not constantly felt throughout early development so embryos begin life suspended in amniotic fluid sort of floating in a state of neutral buoyancy but this doesn't last long as fetuses grow during pregnancy gravity becomes more and more Salient as their mass increases their buoyancy um becomes negative I think until finally the gravity support of their mother's womb is kind of taken away when they're when they're born into a gravity Laden world so babies and children then have to learn to control their body under gravity so mastering all kinds of postures and movements so children strive through effort to approximate that embryonic foundational effortless experience of buoyancy by maintaining increasingly complex states of verticality and balance as effortlessly as possible so this learning involves a lot of falling obviously which is the opposite of buoyancy so there are complex neuroanatomical structures that help children adapt and adopt that sort of complex balance function so the vestibular system which includes the posterior insular cortex regulates verticality through graviception to maintain bodily balance control gate and all the other physiological functions required to optimize bodily states with respect to gravity and so up to this point it seems like we're just talking about physical States right well actually no not even at this foundational level because graviception the sensing of gravity the perception of gravity also impacts higher order cognitive processes related to selfhood so the feeling that we are our bodies that embodied is deeply rooted in the fact that

our body has a center of gravity so the egocentric perspective from which we view the world is anchored in the feeling that we can exert control over a distributed body weight but the mental doesn't actually stop there so it goes to the next level so Daniel so by interacting with physical gravity children also learn to simulate those interactions without feedback from the external environment so you can just imagine yourself moving and the effects of gravity on your body now whereas physical gravity with feedback is mediated by the posterior insular cortex in the vestibular system simulated physical gravity is mediated by the anterior insular cortex and as you may know the anterior insula is part of the salience network that regulates attention interoception effective responses to homeostatic signals and most importantly for mental gravity socioemotional expression so this is how physical gravity states have been co-opted or mental gravity so the anterior insular cortex takes gravity that begins as an external physical and embodied physics and applies it then to the internal interoceptive effective environment you can just click once there Daniel please so hopefully you can see a little pink ring around physical state is that where you're up to yep so that's where the physical states in the socioemotional environment derived from That's How gravitropism has been copied from the external world and applied internally as a template for social emotional meaning and just click once more thanks Daniel or actually twice more but what about the mental States um derived from the internal environment where do they come from so are they just interoceptive like the anterior insula well no that's just the present state of the physical body so Barbie didn't have a fever or have an upset stomach she had an emotional reaction to an event that deviated from her long-term kind of entrenched prize expectations and World model so the internal mental states must reflect long-term states of the self as derived from autobiographical memory and this is where the rubber really hits the road uh for mental gravity next slide thanks Daniel you got to click a couple of times so here we are we're looking at the sort of top to bottom in the middle of the of the figure So Physical gravity states are the present state of the embodied self as regulated by the insular cortex so both posterior and anterior mental States however are not restricted to the present and so are not physically embodied so their source is from the past in the form of autobiographical or narrative sense of self so it's not the what and where kind of embodied self but the more diffuse who sense of self that is disembodied in a personal story or a narrative so in the diagram that's what's represented down the bottom in the internal environment so that's life's ups and downs and thanks to cognitive gravitropism these mental states have adopted the structure of physical States just as it feels good to be physically up and stable in one's body because it has a positive

adaptive value it also feels mentally good to have a higher and more stable sense of self so self-esteem self-worth self-regard so these the bases of mental health and well-being because they are higher more pleasant value States than their opposites so health and happiness are up ill health and unhappiness are down have we just click uh once there please Daniel so when Barbie's life story dips unexpectedly she feels low in terms of selfworth on a scale of verticality value as I call it and then click again please and so she experiences a down state of having fallen and once more and she simultaneously feels low mentally and physically now to this point none of this is counter intuitive or controversial I don't think some of you may be thinking at this point so what this adds nothing explanatory because it doesn't give us sort of a causal mechanism it just describes a process Without Really nailing down how it happens well this is where it gets possibly controversial but definitely counterintuitive this next bit is the stretch point so next slide please Daniel so my radical proposal is that the narrative or remembered sense of self also has a center of gravity just like the physically embodied present sense of self so think of it like this we all agree that the mind has a structure so memories accumulate associate by learning and self-organize into a complex network of nodes and connections well if we take that structure literally and not just metaphorically if we assume by a cognitive gravitropism that the structure has a vertical Dimension so some nodes are higher in the network than others or hierarchy then we can also assume that this structure can collapse so if the network undergoes sort of Rapid or radical reorganization such as when Barbie encounters the shock of negative feelings those high parts of the memory Network structure May no longer be supported by the lower parts so the center of gravity the point at which gravity exerts vertical force on the network will therefore fall and the person will experience a lower mental state than they had enjoyed previously actually don't are we on the soci emotional state is that what the heading is there cool so I realize that that may sound utterly ridiculous on first hearing it's not something we're accustomed to think thinking about so we like abstract conceptualizations of the mind not concrete ones like this we like to imagine the mind is kind of a diffuse ethereal and immaterial object that has no discernable physical properties especially rudimentary physical properties uh like Mass but I think the principle of cognitive gravitropism sort of forces us to change our change our conception of what constitutes the mind so it's obviously non-physical immaterial and purely mental object but that doesn't preclude the possibility that it has analogous physical properties in fact if embodied cognition is true then it makes sense to assume that the disembodied mental World retains some properties that mirror the physical world all right next slide thanks Daniel okay

so let's look at the concept of the center of mental gravity in a bit more detail because it all kind of hinges on this point okay next slide thanks so this isn't a new idea and it's actually not mine so philosopher Dan Dennett in 1992 so he coined the term the self as a center of narrative gravity so he used this as a metaphor but it was a serious metaphor to convey the idea that the self is not a mental object you can place kind of within a memory content but rather a point around which everything else revolves and by everything else he meant autobiographical memories okay here's the analogy so the body center of gravity is not a fixed location it depends on the body's posture and movement well the same goes for the sense of self it's not fixed in one place on one concept it's an abstraction it's a point in mental space and like the body center of gravity it can move depending on how the narrative is changing and what new memories are accumulating around that Central Point okay that's one way to conceive of the center of mental gravity in conceptual terms or philosophical terms another is in neurobiological terms and the default mode Network just a click once please Daniel so more recently Christopher Davey and Ben Harrison from the University of Melbourne here in my hometown so talked about the default mode Network as the brain's Center of gravity so leaning on denn's idea of narrative gravity they observe that the default mode Network represents that narrative or even integrated sense of self so it generates a high order abstract representation of selfhood derived from autobiographical memory so it integrates memory to abstract out the central protagonist the self and they also showed on a neurological level how the default mode Network performs this integration by kind of placing itself at the center of the global connectome Network and just click once there thanks Daniel so in another paper these same authors put forward the idea that the self exists along a dimension from experiential to narrative poles so the experiential self is basically homeostatic States as regulated by the salience network and the narrative self is autobiographical States regulated by the default mode so this connection between those two is very important for mental gravity in associating those gravity simulations performed by the anterior insula to the narrative processes performed by the dmN and next slide thanks Daniel so this idea of centrality of the dmN is backed up by many neuroscientific studies of structural and functional connectivity so along with the cognitive control Network or Central executive Network as I call it nodes of the dmN are the most globally connected regions of Cortex so this places the dmN at the top of cortical organization because the nodes are most dissociated from lower level primary sensory motor cortex and just to click through there Daniel things so in terms of topological organization this means that the dmN is positioned at an extreme of the cical hierarchy because

it processes self-related information that is detached or unrelated to immediate sensory input
it processes long-term trends in memory and narrative and one more click please Daniel so as
argued by Carhart Harrison Friston this placement of the DMN at the top and center of the cortical
hierarchy means it's functionally Central for free energy minimization and the phenomenology
of selfhood the ego and next slide thanks Daniel so if the mind has structure including contents
that are arranged vertically and prone to Falling then it makes sense to also speculate that
memory has a kind of mental Mass upon which gravity is exerted just one click through please
Daniel and so that's where the anterior insular cortex comes back into the picture so it is a key
node of the experiential self because it sits atop the hierarchy of neural processes that extend
from the peripheral nervous system to the brain stem and subcortical regions that making up
the brain's Central autonomic network but remember it also regulates the simulation of gravity
and so the assumption is that simulated graviception acts upon an imaginary mental Mass so
Barbie kind of imagines her mental narrative self falling even though her body is not falling
but then because she has imagined it kind of self-fulfilling prophecy maybe it's self-
evidencing I'm not sure her body then follows suit and she feels physically low so gravity is
part of how the present self interprets information derived from the narrative self in terms of
gravitational homeostasis so remember that embryos begin life floating in amniotic fluid so
that is a foundational state of cooh homeostasis inside the M mother's womb that's referencing
Anna Chukha's work with Nick Wilkinson and like temperature and pH and all other
homeostatic set points or attractors deviations from gravitational homeostasis yield negative
affect so feeling heavy feels bad feeling light feels good so Begins the film floating through
Barbie land but then when she encounters negative affect for the first time she feels heavy and
low physically emotionally not that she had a childhood at all but her narrative has deviated
from that embryonic foundational sense of self so her state of gravitational homeostasis is way
outside her prior distribution but it is not physical gravity that caused her to lie down it was
mental gravity and and next slide thanks Daniel okay so that brings us to the second last
section so the relation to active inference so that idea of violated gravity PRI brings us to um
the free energy principle and active inference so I'm not an expert at all but I think there are
some connections that we that can be made between mental gravity and themes within the
active inference field and next slide thanks so the tenets of mental gravity map nicely onto the
basic puristic of active inference so just one click there please Daniel so external States would
map onto the bodily interactions with the external gravitational environment centered on the
vestibular system and posterior insular cortex and just one click through sensory States would

map onto the activity of the anterior insular cortex the experiential and embodied self which combines homeostasis and affect with simulated graviception and next click thank you so internal States would map onto the default mode activity so the inner workings of the mental narrative self with its Center of mental gravity and one more click and active States would be the combination of sensory and internal States via social emotional expression in thought language and behavior so they're the descriptive similarities but I think there are deeper conceptual parallels as well relating to that radical idea of a center Center of gravity and next slide thanks Daniel yes yeah thank you so this study by Kang et al exemplifies what I mean in free energy terms about the centrality of the dmnn so they looked at energy Landscapes of brain States using um estimating using pair pairwise maximum entropy of resting state fmri so they identified multiple local Minima some of which mediate multi-step transitions towards the global minimum so what is the global minimum so it says here co-activation of the prefrontal cortex and default mode network is captured as the global minimum so as well as being Central the co-activation of the internally oriented dmnn and externally oriented Central executive Network kind of represents the Baseline of the brain's energy landscape so in healthy or unpleasant States healthy or Pleasant States the dmnn and Central executive Network act together to maintain a sort of stable mental Baseline but what happens when that Baseline becomes unstable or moves away from a homeostatic set point in pathological or painful States next slide thanks Daniel so a recent study by Friston and colleagues demonstrated that in depression when I would argue the Baseline kind of drops there is a shift away from the central executive Network and Def towards the default mode and salience networks so in conjunction with sensory networks and the Cerebellum so the self has become even more Central in an unbalanced way that people experience kind of as a fall away from that external world and into themselves so rather than feeling connected to the World by a sort of permeable boundary it is as though their sense of self has become their entire world so their past becomes all that there is the future out there recedes away from a never ending past in here that they cycle through in rumination so next slide thanks Daniel so that's what's depicted here in this figure in my earlier paper on mental gravity with reference to the Continuum of mental gravity States so up the top of the figure on the right the self-world boundary is kind of where it should be in positive up or down States like joy and relaxation so the salience network is paying at least some attention to the world and so there or directing attention to the world and so there is a clear distinction between self and non-self in depression at the bottom however when people feel down and low on account of a fallen sense of self there's a sense in which the

increased centrality of the dmN narrative self enlarges kind of the weight felt at their Center of mental gravity which then pulls the salience network which is kind of at the boundary down and within that self-world boundary so anxiety is included in that diagram in the middle but I'm skipping over that at the moment so now gravity in physics is equivalent to space-time curvature so Mass distorts space and time so I make the case in this paper that this literal depression of the mental self is akin to space time curvature across all levels of phenomenological cognitive and neural Dynamics next slide thanks Daniel now this is where for me it gets very interesting for active inference and the free energy principle because the physics of SpaceTime the general theory of relativity can be considered part of the gauge Theory framework as argued by Sen Gupta et al in this paper from a few years back so mental gravity if considered a form of curvature of neural cognitive and phenomenological SpaceTime could also be integrated within this gauge Theory uh neuronal gauge Theory framework just one click so consider this quote from their paper so crucially the cognitive homologue of precision weighting in attention is ATT tension which suggests suggests gauge uh fields are intimately related to exogenous attention in other words attention is a force that manifests from the curvature of information geometry in exactly the same way that gravity is Manifest when the SpaceTime Continuum is curved by Massive bodies so for me there's a clear connection between mental gravity as a principle that connects external and internal environments via social emotional expression and mathematical Frameworks underpinning active inference that are already kind of flirting with the analogy between space-time curvature in physical gravity and psychological Properties or processes and next slide thanks Daniel so as an aside before I finish up I also think there is a way to implement space-time curvature of general relativity in the brain in a way that is consistent with the tenants of mental gravity and SpaceTime curvature of cognitive and phenomenological Fields especially for depression so this is a work in progress with Dennis Leary we don't agree on everything and he certainly doesn't agree um that mental gravity is implicated here but he he's the author of a relativistic model of the SpaceTime connect time so there's more to come on this we'll keep working and that's kind of it that's a lightning tour of mental gravity so my theory of kind of what we are made for so the main objective is to provide a model of mental well-being of being up and stable this is what I believe we are made for um according to gravitropism and gravitational homeostasis so you call it what you like human flourishing happiness meaning in life however you want to phrase it so that's it thanks all right thank you and thank you for to you and the audience for bearing with some slide fun as we discovered the minutes before

beginning but well done okay well people watching live can add any questions but uh while they are adding any questions in the chat just how did you get to this project how did you get to active inference what just kind of LED you to being here now so originally the connection happened sort of in my undergrad I was studying psych and physics at the same time when we were doing um sort of black hole physics in astronomy and astrophysics and we were talking about depressions in SpaceTime and then an hour later we were talking about you know depressions in Psychopathology and I think it was really a linguistic thing that got me first I'm oh okay we're talking about very similar things here that people seem to feel down and dark and slow and heavy and we've just been modeling that mathematically in physics I wonder if there's a way to use that maths to model something about about brain structure or function and then during my PhD I connected with yakob hovi here in uh Melbourne who's doing self-evidencing and I think studied under Carl back in the day and so he introduced me to active inference and predictive processing we published a on that as part of my PhD and so I started thinking about okay there's something about prior and and you know Human Experience of gravity that is probably expressed somehow in mental States in mental well-being and that's kind of what draw me drew me to it all right cool well I look forward to any questions in the chat but I just kind of wanted to reiterate what I thought was one of the core points POS Ure is basic to moving animals and to mammals and you pointed out that that gravitational homeostasis is like pH or nutrition or sleep it's a fundamental moving set point situation if you want to get over a small fence you might want to briefly have the distribution of your physical center of gravity move up you might like plan out different routes just like you might plan for your or um implicitly or explicitly like your glucose levels to fluctuate and so that kind of like skillful competency definitely connects to the fields and leaven work in the competency and generalized State spaces so then that kind of navigation plus gravitational homeostasis becomes co-opted analogously to how the linguistic does like evidenced by your most recent answer but the gravitational homeostasis becomes co-opted as part of navigational which is as a cognitive or narrative degrees of freedom in generalized State spaces and there's just so many ways to go and it was awesome how you laid it all out so again thank you for presenting thank you where do we go from the center of gravity basian mechanics what kinds of older data sets theories do you think are like worth revisiting or what kinds of approaches does it open up um I've worked with uh Paul Badcock who works on the hierarchically mechanistic mind as well he's in Melbourne here so he's worked with Carl and others to Define sort of evolutionary and developmental nested time scales um inference and I think that you that kind

of nesting of present experience within developmental experience within I don't know cultural and evolutionary experience I think is a way to um way to approach it so gorg norof who was maybe going to join us here today but hasn't been able to make it so he would talk about nested time scales in even a a theory of Consciousness that that nestedness is the necessary part of necessary part of experience and deviations from nestedness lead to pain so those homeostatic set points are nested um so I think there's work to be done there in the experience of time is really where it started for me the experience of time dilation in depression to me is analogous to gravitational time dilation so masses curve SpaceTime so that time slows down there's very good empirical evidence that in depression people experience a slowing of their conscious experience so not it's not like their internal clock is running slowly it's not like they mis judge times all the time I think there are really fine technical ways to elicit changes or distortions in their time perception but their subjective flow of conscious experience definitely slows down so in Mania the opposite of depression that experience of time speeds up in anxiety another Upstate it's also sped up kind of chaotically so I think there are ways to I think pull out okay what's your experience of time and then how do we sort of recover that how do we re-nest that through whether it's postural or I think maybe better to have you know mindfulness meditations meditation that uses simulated gravity so asking people to imagine themselves floating so I do a lot of yoga and part of my practice is Imagining the body lights that's all postural and it's trying to imagine all movements a bit like Tai Chi that there's there's no weight you're just moving freely and so that seems to me to improve mood I think because that center of gravity the physiological center of gravity has then fed back on the emotional state on the The Narrative sense of self because they're connected the anterior insula doing that simulation it's connected to the default mode and there's some feedback there so I think time and sort of perceived imagined body weight are two ways to sort of attack the problem all right awesome all go to a question in live chat upcycle Club writes how does the concept of mental gravity influence the perception of time and space in astronauts versus normal folk okay that's pretty complex so because they go from a state of you know regular gravity to micro gravity in space and then back again so Studies have been done showing that there's um sort of visceral changes visceral and physiological changes in space that lead to um it's not again it's not a change in perception of time but there's or space so the perceptual field remains pretty um unchanged but when people come back to earth and they they sort of re-acclimatizing to gravity there is a tendency to experience anxiety and depression now that could be related to the fact that they've just left the space station and they're back on Earth and

they're feeling a bit you know bit let down by it but there's there's ongoing work happening that um I think they've just done actually some psychophysical experiments with the perception of time and they're not noticing I don't think anything um major other than sort of the sensory motor stuff so people are having to like when babies are born they have to recalibrate how they move for Gravity there's a an adjustment period where people have to um reac climatize and re sort of retrain the sensory motor Network to let go of the gravity expectations I'm not actually hugely conversant in that that's somewhere that I need to learn more about because the work that's been done yeah it's it's it's formative but yeah we're getting there cool one thing that that mention of the microgravity and and the call back to the developmental part made me think of and you referenced and C unica's work then is is like in the Single Cell water environment there's a neutral buoyancy so that means that all so all subsequent gravitational homeostasis can be attributed to self and movement whereas um nutritional homeostasis has a weaning off of and never a complete weaning off of but gravitational homeostasis really remain me but also from the very beginning as something that um it can be inferred reliably over evolutionary time scales that it is under that mammal's agency yeah exactly yeah it's a universal constant and we're now pushing the boundaries of it by getting ourselves into microgravity where you know no animal other than I suppose fish you know get neutral buoyancy and I think there's an argument that Nick and Anna make in their paper on sort of buoyancy in utero that what we're trying to do all the time is approximate neutral buoyancy with posture and movement that it's the feeling of load that is a deviation from that ideal State that's the the prediction area era that we're trying to minimize is the feeling of load and there's another uh there's a person in America who's a gastroenterologist who's working on gravity in the gut and linking that to mental health conditions like anxiety and depression through irritable bow syndrome and his idea is that sort of a failure of the gravity support mechanisms in the gut leads to you know changes in the gut microbiome that we then interpret as you know a change in G forces it's like the gut is uh a gForce Accel accelerometer a bit like your vestibular system senses changes in the head the gut also has you know propri acceptive visceral um sensors for acceleration and that's why we say we when we're nervous or anxious we've got butterflies in the stomach it's like the the gut has become light and that's from top down predictions a bit like Barbie saying I'm I've had this fall and now I'm lying down your expectation when you're nervous is that you're under threat and that there's some change in the gravity um acceleration that you're feeling in your gut so here and I in conversation so talking about how the gut brain connection could be you know

another way of exploring mental gravity as you know not just brain and nervous system um your central nervous system but also your peripheral nervous system and and visceral feedback and and that works for predictive processing because Inter reception is very top down um dependent wow makes me think of yes butterflies in the stomach and also heart or stomach sank like a stone sank Like a Rock which is to say like the prior experience no resistance under the constraints of normal free falling things it went down at this acceleration and uh to ask about that linguistic angle like are there any metaphors that aren't consistent it seems like in day-to-day conversation it would almost be taken as a strange metaphor if something were inserted kind of like Kandinsky who asserted that warmer colors were associated with light and with higher notes of music and like cooler slower darker and lower notes and it was like well if that when violated surprises people then that speaks to the nature of that associative prior but I mean are there any linguistic examples or anything that not that I've come across no because I I assume that it's preverbal that children have um you know natural associations between higher space and you know certain characteristics so more we have a metaphor that more is up so we're saying you know greater value higher value is more things and babies will attend to higher space when there are fewer things than um when there are lower so that violates their expectation they expect more detail in higher space so it's yeah I think it's Universal across in English across languages and because it predates language so I'd be very surprised if there was a contradictory metaphor the one that there are behavioral um Expressions that are kind of contradictory so I think of when people say win uh a tournament so they've exerted all their energy and they win often they'll fall down they'll collapse in sort of joy and so that's kind of a behavioral contradiction so I think taking it too simplistically is going to be a problem because you know counter examples will be easily found but I think there are ways to frame that in terms of mental gravity it's not just it's a very complex interaction with the gravitational environment so lying down feels good sometimes so what they're expressing or what they're experiencing emotionally at that point is kind of being overwhelmed by the volume of emotion that they're feeling so it's still an expression of something gravitational it's just kind of inverted so I don't want to be too simplistic about it and say up is always good balance is always good because in play in Risk moderate risk taking falling is good so we've got to be very nuanced about the language that we use yeah great all right question from Dean tickles the response to being embedded in a gravitational influence setting seems to be quite distinct from being embodied so what does a buoyant mindset look like always good question mark yeah I think it always feels good but whether that's always so

homeostatically feels good whether allostatic that is is always good I'm not sure so if alisasas is about the prediction and anticipation I always think of sort of MDMA at this point so there's a tradeoff so people will take MDMA to get high and to have a super ecstatic super buoyant it's almost like gravity doesn't exist they can dance for hours but then two days later they're going to hit the wall and have a huge come down so I think it feels good in the moment because that's the anterior insula homeostatically saying this is great this is cool but then the dmN is maybe going to anticipate and say this isn't going to be great for our narrative there's going to be a longer term uh cost to this experience as well so yeah it might feel good in the present but that doesn't mean it's going to feel good in the long term then we've got Anna joining us have we yes yeah did that answer the question though did that get to the the heart of that person's question I I hope so yeah cool and are you here hey hi Anna how you going I'm okay how are you well thank it's very late here I don't know what time is there but it's like it's Sunday evening we're here we're in a three sector conversation which is uh you know equidistant but also some of the most fun to coordinate as we know yeah yeah yeah yeah um do you want to like say anything or ask anything Anna or or we can continue talking or reading other questions no no no I don't I I don't have questions for now so good okay cool yeah another thought or area was you talked about up and down but under a closed Planet structural model up is out and down is in it is gravity so it doesn't it's that we can have a meter stick coming perpendicular or normal and talk about up down in a reference point but isn't that the whole question about whether you're up is my up and whether your toilet flushes the same way and all these like questions like that is your left my right all these things yeah so I think for like neurologically the representation of out is into the external environment so that's the central executive Network that's the main mediator there and so we're connecting with the external world that way our default mode is our internal world but we also have to imagine what other people have inside their internal world so there's you know what's someone else's narrative so that has to then connect with other people's up and down so I think being able to then so children start very egocentric and very you know they're not mind readers at that point so then to be able to not only disembodiment their own sense of gravity or their own Center of narrative gravity to then apply that to other people involves another turn another okay that other person is out there and they have an internal World um as well how do I engage with that and that takes then behavioral coordination in the external world so behavioral control empathy and self-control or compassion and self-control go very much together so in that sort of relative space of multiple Minds multiple centers I'm just centered in my own but I need to

connect with others I need to be able to regulate my behavior if I'm manic and self-interested and every everyone's up is my up then that's dysfunctional I think or socially and emotionally not going to connect with people well that's why we kind of term it I don't like saying pathological but that's why we term it as socially dysfunctional and unsustainable in a way it's not they're not embedding themselves in that social environment in a way that acknowledges that other people have different orientations that their up might be my down just to kind of like echo part of that um when there are Polar Opposites it's a polarized antipodal situation um one's up as the other one's down but even someone standing relatively next to you has a partially diverging perspective except when there's the construction of the flat playing field like an ice rink or something that's been then secondarily engineered to be even flatter than the local curvature like a linear accelerator for particles is extra flat which allows extra measurement accuracy coordination but then as it goes further and further out it becomes like a larger and larger project to maintain all of that yeah [Music] Anna yeah so I since I uh I read your paper L is um I said next next time I'm going to see you I'm going to nag you with a question okay about this because um I wonder how much of this like um narrative between you know up is light and good and down is bad so low it's coming from this like inherited um you know inherited um ontology where you know the light and God is somewhere in the you know the sky and uh the devil and the hell is in the Earth right is down yeah so basically we want to escape the hell and the the the death because we are all going to die and you know Earth to Earth ashes to ashes you know yeah we want to just like go up and uh and that's that's good in that sense right because we well nobody wants to die and I think I I love that because um you know um p deola and all this Renaissance people um when they were fighting against uh the church was like stating that the Earth was in the middle of the universe and actually everything was revolving around Earth and they were refusing the idea that you know um actually is the sun is not the Earth and then they were saying well if the hell is in the middle of the Earth and everything revolves the around the earth so everything revolves around the hell the universe that kind of like ring the bell ding ding ding you know the heads of those people like higher up in this like I think from now moment on I kind of like realize that okay so maybe there is something about the sun which is like you know the Heat and the but my question to you is like how much of this do you think is kind of like related because in in in a sense like the food is coming from the D I mean the food is coming from the earth right so everything that actually allows us to survive it's not so it's so don't you think that somehow this like we have like some side effect of some very old ontology that we inherited where you

know God is in the sky and it's good and it's Eternal and death is here and it's bad right with the Earth but that's question because I said it's like if we don't have the carrot and the carrot is not coming from the sky it's coming from underneath that right so how how you reconcile that so I think like being absolutist about it is a problem and saying that up is always good no matter what down is always bad no matter what I think even in our Behavior or even in like a mature approach to life a balance of up and down is required you don't want to be happy all the time there's no adaptive value in that you have to go through trials and tribulations so a kind of childlike I want to grow up and I just want to go to the stars and you know be Superlight and be sort of divine I think is kind of you know counter counterproductive in a way and I think like we have words of being grounded and of being mindful that are very down oriented so I think there is a value in things that are here that are stable that are um connected that are in Balance like a lot of environmentalism is about Humanity in balance with nature so that's possibly even more ontologically prior than monotheism and the idea that you know everything up is good yeah I I agree with you but you know it's just like when um when I'm very tired right like for instance right now just like lying down is actually goodly want to stop and lay down I don't want to go up right so so it's it's a as you said it's like it's balance yeah and we need to be careful not to smuggle in a sneak in without realizing some like from the kind of like you know dualist ontologies like we have God and the things good in the sky and bad things are here which is yeah yeah yeah that's just like a coping mechanism is I don't think that's that holds because we actually need to lay down yeah exactly actually half of our life is like we're lying down because you know we need to sleep yeah so I think there's a kind of a way out of that sort of paradox and like in monotheistic religions they're talking about the afterlife in a sense like that's like if you had a bird's eye view of your entire life it is if you'd never gotten up if you'd never been vertical you wouldn't necessarily be you know classified as having lived so it's a necessary condition to get up but it's not sufficient it's not doesn't suffice the whole you know all the contents of that life so you can say well if the religious people want to talk about that then that's an afterlife reflection it's not a lived embodied way to be that is more about balance and the complex like we were saying the complex nuanced you know if I fall down that can it's like you can cry from laughter and tears you can fall down from depression and ecstatic Joy so it's about understanding those those complexities that makes it real world and not mythical so yeah in the paper or I can't remember which one I said this is what mental gravity is not it's not about the fall of Man it's not about you know some mythical sense it's not about quantum gravity it's just about def

trying to Define and model all our complex social emotional states using gravity as that prior experience template yeah I'll ask a question from the live chat um upcycle Club writes in terms of embodied information processing the brain processes information at different granularities how does this impact our conscious experience of mental gravity well I look at uh sort of spatio temporal granularity there so spao temporal scale so I think a lot about time and the nestedness of temporal scales so at the lower level you've got fine grain temporal scales so fast time scales and that would be for me analogous to the anterior insula Network that has been shown to be very much related to the perception of faster time scales at around 125 milliseconds so it's very present oriented so that is one level of granularity the default mode network is much more a granular so it's much more thinking about longer term Trends and for mental gravity so when there's when you've got centrality you've got nestedness of those time scales they're sort of able to harmonize with one another even through you know neural um oscillations like you've got that's G norof northop's work which is looking at spao temporal alignment of those time scales so that's when you're feeling good and you're feeling not overly centralized I think that's when the power of the dmN becomes too overweighted a healthy or you know a light state of being is when there is alignment between spao temporal scales so yeah it's the difference it's the connection so if there's a dimension of the self from anterior insula and I should do it this way anterior insula to default mode then the nestedness of those scales is very important wow very interesting on like a little bit less of a neurobiological note that makes me think about estimating time estimating like double click speed all the way on through seconds and minutes and hours and days and you know there's decades to take weeks and weeks to take decades all that kind of discussion and a day on Earth is like this long in this other thought and then those kinds of uh compression and dilation are used to point towards the direction of a distribution it's kind of like saying yeah from this GPS coordinate to there you go this many East and that many North and then similarly in cognitive or narrative spaces like oh I was looking for this kind of album but a little bit more of that and from like this period or with faster beats yeah exactly so that kind of like rate and gravity connection it's also so concordant yeah it's it's very clear and there's Paul Badcock's work so the hierarchically mechanistic mind with who did work with Carl so he's also done work on um the social adaption theory of depression so there's mapping there between his thoughts about the nesting of sort of evolutionary time scales through to you know ontogeny um and then fast neural time scales um and the disturbance of that in depression so the social and emotional disturbance of that alignment in depression that you depression is a way of sort of

calling out to other people that you need help so Barbie is saying I need other people I'm not just happy in myself I'm calling out to others I need support to get myself vertical again and yeah there's a nice Synergy there with Paul's work as well yeah that's awesome I was thinking of a few other like you mentioned a person collapsing at the end of a sporting event and how that's a combination of like muscular fatigue and the desire to have the homeostatic set point B down because they want to be resting closer to sleep all these different things or the symbolic meaning of it and and all that and then I thought about the float tank experience and um fun fun house rides uh amusement park rides like spinning and uh and the dipping and then weightlifting as kind of an integral Through Time potentially reducing the perceived weight load yeah I mean so how would you just say at this at these early days for the um the Gravity Research program like how do you take a phenomena and just sort of as you've probably done many times like try to work through it in this gravitational way I think multi multidisciplinary approaches is the only way to go so got the guy in America doing gut brain stuff so gut um gravity there's also Mark Whitman who I uh published a paper with on Consciousness on time Consciousness he's very much into flotation tanks and using that to alleviate anxiety and depression I also think that psychedelics there's something about going on a trip there's the phenomenology of you know expansive space and time that happens within um Asos cyber trip that we can talk about that is the therapeutic angle is something about SpaceTime curvature or um the reentering of self an altered state of self altered state of consciousness so I think there are multiple ways to look at it so you know different interventions different conditions and different sort of emotional states that we're trying to um trying to model without you know forcing people into doing weightlifting and saying it's all about this one monolithic way of doing it like Anna was saying like it's not just about sacred spaces that are tend to be expansive as well so churches and larger spaces even forests natural blue and green spaces have that expanse to them that's sort of um therapeutic in a way I think there are multiple multiple ways of looking at it it's like I suppose it's like predictive processing that's a way of looking at things that you can apply to different fields I think mental gravity is something like that it's very foundational it's kind of um Central and implicit in many fields of work so I'm hoping to work with Mark on flotation with um Brennan in America on the gut brain stuff with Anna about and Nick on you know cooh homeostasis and the development of graviception In Utero I think there are many many ways to look at it awesome yeah Anna thank you so much I this is really really interesting um I have a question related to the to the fact that um so um I understand that verticality is super important but um don't you think that

is also that connected verticality is connected with the fact with The Locomotion with the fact that people can when they are vertical they can basically just like you know go there in the world and find food you know self and make sure sure that and that's that's it's not necessarily because it's not enough just to stand right no yeah vertical it's like if you're still standing vertically is not going to get you very far the same way if you stay still like laying down it's it's all the same right um so um how you see this like verticality in relation to being static and being um active so I think the active component is like related to motivational control so explore and exploit behaviors so being able to explore if you're vertical you have greater degrees of freedom of movement than when you're lying down so it's easier to get horizontal if you're vertical but you also need a low center of gravity need to maintain a a balanced uh center of gravity in order to do that so more agility more ability to move side to side is about maintaining a low center of gravity when you're vertical so they're kind of working in combination so I spoke to Nick Wilkinson about this and he's like okay it's when you're born you're starting at this I can't do anything I I can't move I can't get myself up I'm dependent on my Caregivers for all exploration of the environment I can only see I can't interact yet then over time as we develop that motivation to explore comes along with our ability to maintain verticality and prevent fall so we start by Falling a lot but that's because we haven't learned that that low center of gravity is what we need to regulate and I think it's the same for mental gravity you need to be vertical but maintain a low center of gravity when you're vertical that's most sustainable and better for like exploring yourself exploring ing you know your own narrative if you're high and unstable you may might be in a you know manic or anxious or you know overly Upstate that's not adaptive either so there's I wouldn't want to say it's a u curve that it's you know it's there there's an Optimum state in between because it's really complex but it's a bit like that there's a a combinatorial way of approaching gravity that means it's not just about verticality and it's not just about low verticality or misalignment about horizontal it's about a combination dynamics of of the two that that's an amazing point and I think points to kind of a key engineering trade-off which is the bigger the building taller the building more further out the building goes the higher you can see and then the higher the center of gravity becomes which then under even small shaking of the land or like Tower of Pisa status that becomes comes unstable so then when skyscrapers are built there's a heavy weight to move the center of gravity down and to dampen all these kinds of things that allows like taller narrative construction so that's super interesting and then I thought about insects and not just flying insects because there the verticality is not associated with sleep like an ant nestmate will be

active at a given elevation and then stop and rest but they're not going down whereas mammals even ones that are on quadruped um Locomotion they still do go down when they're cowering or like curling up but in the insects they they can't go well you know outside of the ones that like do and all the stuff like that but just in general a ground walking insect doesn't go down so that really isolates out the affordances and kind of possibilities for movement because I think Anna made a great point that humans the verticality becomes also associated with like being able to move basically so being down it's like you're caught on flat feet it's not even just being standing you have to be like in a little bit more of a poised posture to react to basically anything even to catch something very light like thrown at you so that's very interesting how the phylogenetic priors which go far deeper than any of our um at least symbolically describable or understood cultural priors which also are very relevant and recent but these kind of batara or going on land or having aqueous components those are even deeper and like you said they're preverbal and then yeah for sure I yeah yeah I think this version of mental gravity is uniquely human I think I don't know if we can generalize it I think there might be different homeostatic ways for those animals of talking about it but I think this you know our up down associations I think a preverbal sort of you know non you know they pan cultural but I don't think they're inter species necessarily I think it maybe for fish there's something about neutral buoyancy as well but for insects for the birds who knows but that might be one of the characteristics of their space your temporal Consciousness who knows yeah yeah like for a fish there's a possibly strong environmental correlation between like the density and the temperature and the light and so there's a affordance in principle to move up but then they find themselves in a niche that might be like seasonal but still captures this kind of like density light nutrient envelope because that's their Niche yeah yeah oh yeah ecology it really is like that and about finding the right level of you know fly when it's time to fly but then not fly when it's not time to fly all that yeah so you could you can do experiments with that in you know fishing a tank that you accelerate you know to different g-forces and see how they behave differently and what associations they make with nutrition or not just buoyancy but you know light like you're saying there're probably prior from you know evolutionary prize phenotypic um phylogenetic um prize in there that you can elucidate yeah wow that's very interesting and bacteria or other single cell in even just small tubes there could be gradient centrifuge with with constructed nutrient gradients that's awesome experiment um on the active inference side um you showed the particular partition with the internal and external and the sense and the action States the four states so the three states the sense internal and

action are the ones that are defining the agent or the or or the organism and so I was thinking like what is a heavy sensory experience well heavy metal a heavy cake something that and then what's a heavy action possibly it doesn't need to be even any more literal than um heavy deadlift or it could be a very burdensome task that's hard to to push and then heavy internal States so it almost has a heavy and light not veilance axis but just in this kind of mul dimensional space of active inference agents and models there is kind of a separate heavy and light for these three different parts of the particular thing so I think that also just speaks to the applicability it's not saying that those three states of the thing are like the same because they have to be welded together by mental gravity it's actually that gravity is like an analogy frame or a metaphor that at the very least can be used and is used normally and you could also invent speech things and just see which which one sounds more normal which one sounds more like the language model wrote it and just used never seen before gravitational language versus anti-gravitational language just see what the effect is maybe people prefer one or the other if it's like fantasy well there there's work been done at my University at rmit in Melbourne by someone who used Pro so the upward and downward inflection of speech so pitch modulation to analyze um speech of depressed and non-depressed people as a kind of biomarker so how people talk the naturalness of their Pro is kind of could be Diagnostic in a way because it shows what sort of emotional veilance is being used when they talk up and down there's really a flatness of tone in depression that people can sense and people are expressing I think because of this like you're saying what what's the metaphor it's kind of like cognitive load we think about depression as you know the self has become too Salient there's too much self-referential information that's a drain on cognitive capacity because it in itself is cognitive load so their ability then to think about the external world is limited so I think there are again multiple ways to look at this and multiple ways to test it and intervene and you know for therapeutic benefit so yeah I think Pro as a what what's the weight of Behavioral expression I think there's posture there's facial expression there's Pro there's hes of wise looking at active States as well so then to connect it to writing first off it's very fascinating that language models can through the statistics of language without using things like the pro and the timing just from transcripts learn so much about generating other realistic sounding transcripts um so it's like what information is in the procity and the timing that's just very interesting but it can be a very I mean it helps us think about what those kind of communicative techniques are yeah and the symbolic basis of language is more of a latecomer than the kind of resonant nature of communication on these types of um on that level on that vertical Dimension I suppose you'd

call it yeah yeah yeah and so Nick Wilkinson who's worked with Anna talks you know he's very interested in working on sort of Soft Robotics so how do we train uh an agent not based on um you know a desired state that we train forward but based on like that co- homeostatic neutral buoyancy that's the set point that's the attractor that you're deviating from and then all the implementations of that regardless of physiology is translatable then so that what you're trying to approximate with standing you know quadrupedally is neutral buoyancy but supported with different structures and functions so I think there there's big application there not just in terms of language models but in terms of Robotics as well yeah all of the features of embodiment in environments that we know like if it was a muddy environment or if it was a slippery one whatever the embodiment were saying you have to stay upright in this environment and move is kind of enough of a forance to test and explore the features of that landscape if it has like gravel or what whatever situation it has that's kind of what it has to come back to otherwise like you said to to to get up no more is to die yeah exactly yeah super um interesting Anna do you have any last comments or questions yes so I am I do um uh so first of all I want to comment very quickly when you just said uh L because I think this is super important so um there is right now we are in the middle of like some sort of like like shift in the perspective and I think it's for the best in the sense like instead of starting with something highly sophisticated from the adult Centric perspective we look at basically how or going get there yeah and then we see that basically as I said well um the the body needs to be like you know um embedded into environment in a certain um states that have been designed by billions of years Evolution right so it's not it's not propositional attitude content that we can kind of like logically extract from some abstract formula yeah um it's it's something that is done through trial and error and actually this is something that changes all the time because while you change environment you need also to change posture and who know it's like with all the pollution that we have right now in the world maybe we're going to adapt and develop new organs of s yeah uh so so this that's um I think that's that's something very important but um I I really like the paper and I really like the the way you you know um bring the like the um the basis back in this in the core right like uh in in under the spotlight but my question is to you um so uh that that's that's very beautiful way to um understand like psycho pathology in terms of like uh you know heaviness and I was actually thinking that light and light means both light in sense like not heavy but light like in sense like you know dark not dark right so that's that's also beautiful but um so how you include basically the social component in it so the role of the other in terms of like this Dynamic of the gravity because we um um interestingly enough

people who are depressed or something they disengage also from social interaction so it's like some sort of like flatness uh not uh and heaviness but um not just heaviness but also like disengagement from the active exploration of environment be it physical like you know going to the gym or social like going to meet friends yeah so what is the role of the other in your model for this like um up and down um uh a structure so I think that I would go back to Paul Badcock's work about the social risk hypothesis of depression that it's a bit like okay you take MDMA to have an extreme High knowing that two days later you're going to have an extreme low I think his argument if I have it correctly is that depression is kind of similar that it's it's actually a call out to other people to say I'm going to be horizontal I'm laid out I don't have any energy I need you you to help me bootstrap and get back to being vertical but that can become pathological not by any sort of neural Dynamics it's not a brain disorder in that sense it's a social disorder because that care is not forthcoming or it's not um it's not appropriate or it's not effective so I mean we look at a lot of the social determinants of mental ill health that's the field that I work in really day to day is how do we promote and promote mental well-being and prevent mental health conditions by uh mod modifying risk and protective factors so that risk that I suppose the depressed person is taking is they're saying I'm the only hope that I have left I have none for myself is that someone else will come and bring me back into connection with the world I can't do it I've run out of energy it's like a the idea that um depression is failed anxiety there's an anxious phase prior to depression of reaching for connections reaching for stability and then when that has failed then it's like I I have nothing left so I think it it's a way of depathologizing the individual destigmatizing individual and looking to their environment to say how is society how are other people letting this person down how have they been let down by the world around them so I think it naturally implicates others and take even though the individual is focused on the self the effort from you know a therapeutic angle needs to be from the external whether it's the natural you know uh green and blue spaces can be therapeutic to connect to the natural world or connect to the social world yeah that's that's super important thank you and not necessarily sometimes we let people down but in most of the case also we put people down you know like put me down H yeah and let me down the the other can be for the better and for the worst right yeah also on the gravity and the multi player multi-agent strange attractors so there's like the three body problem binary stars all these asteroid belts all these different stable configurations or unstable configurations or chaotic or rhythmic all those different movements and then on the perspectival part there's the world at the center self at the center ego Centric view which is our kind of undeniable object

related visual system and then there's the kind of other centrism like the sun everyone on the earth will reference the Sun as the center point or there's kind of in principle like a geocentrism but that's always alluded to as like kind of in principle true but then it doesn't ever license any kind of perspectival analysis except hypocritically because a sensor has to be chosen even if it's like a purely homogeneous gas then the center of gravity is just the center of the space and so it's like there's so much there with um how diffused the material is and then well the gas clouds that are really far away they have centers of gravity so just because the Mist is very ephemeral at the time scale that we're moving doesn't mean that for like a redwood tree or for something like a bacterium it doesn't have a totally different materials property so we can't really say like how the niche is by itself it's not like the desert is just hot because it still might be too cold for some other adjacently living creature yeah exactly so there are me there are gravity models of Economics as well so it's not just like we're talking we're extending the physics of gravity for the first time into mental the mental domain they've been using gravity models it's Newtonian gravity it's how sort of cities or states trade with each other where you're analogizing Mass to the size of the population and then the transfer of force is the transfer of goods or services or people or whatever so I think there's a way to apply this at different scales as well to say okay if Society is coherent and nested and you know based on sort of topographical or topological features of that Society you can start to say well where are people going to fall down in this area here is where you're going to see an increase in anxiety and depression which is what we're seeing now we're seeing okay we're using metaphors like Society is fracturing or you know there's this over centralization of power in you know the top you know the 1% of the population so the people at the bottom are struggling so I think there are ways to also take this the mental gravity idea and apply it at different social or societal scales as well awesome yeah well Anna if you have any last comments or questions and then otherwise you can have the last word on like where we go or what do we do but Anna first then Ilin no I don't I don't have any last words so I'm going to let back last thanks for joining Anna thank you so I think um to sum up I think okay mental gravity this is one um aspect of sort of bio gravitation that we can look at I think the conversations that I'm having with people sort of around the world in different uh domains in different fields there is a way to make I know kind of a discipline out of this or at least a network of people who are interested in the effects of gravity on whatever it is whether it's the gut whether it's the brain whether it's development whether it's Psychopathology Consciousness whatever so my call out is to people if you're interested and if you want to

make contact I am thinking of trying to sort of become a node in that Network and try and um bring people together people like Nick Anna is interested given his work he's going into robotics or he's interested in going into robotics so I think there are many different fields that can come together to say okay we're all talking about the same thing how can we sort of formalize it get some you know agreement and some principles and all be working from you know at least a point of point of connection action and discussion so that's hopefully where I'd like to see this go not just in theoretical you know advancement but in socially I think we can bring some of these ideas together and Daniel thank you for having me thank you for contacting this has been great it's forced me to really get my ideas in order and be able to communicate them so thank you for the opportunity and to the active inference Institute thank you thank you up up up yep up up up all right thank you see you all next time thank you bye